

Twenty-five conjectures from a local–global random model, with machine verification and adversarial audit

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Abstract

We state twenty-five conjectures in elementary and analytic number theory, each derived from the local–global random model of the primes: Cramér’s model corrected by Hardy–Littlewood singular series, with Bateman–Horn as the organizing framework and Borel–Cantelli accounting for sparse events. Each statement is calibrated to the strongest form the heuristic supports and no stronger. Every computable constant is computed from its definition, with local admissibility asserted programmatically; every statement was tested against exact counts, with success measured by the *shape* of agreement (predicted constant, square-root residuals, no drift) rather than by the size of the range searched. The statements were then subjected to a three-layer adversarial audit: an independent literature search for prior art, scripted refutation attempts a decade beyond the original bounds, and clean-context re-derivations of constants and counts. The audit refuted one conjecture as first stated—the exceptional set for $n = p + k^3$ contains the infinite family of cubes k^3 with $3k^2 - 3k + 1$ composite, an algebraic obstruction the probabilistic accounting missed—and it appears here restated, with the obstruction promoted to a theorem and the original error documented. We are deliberately conservative about novelty, and separate two kinds of content. Most of the twenty-five statements form a *benchmark suite*: fifteen are previously stated in essence and are attributed, and several more are unrecorded but mechanical specializations of Bateman–Horn—bibliographically new at most, and offered as calibrated test data rather than as new mathematics. A small core carries what we argue is conceptual novelty: a quantified Chebyshev-type race among twin-prime pairs modulo 5 whose deterministic term is driven entirely by the pairs $(q^2 - 2, q^2)$; a finite-modulus deficit law for least primes in arithmetic progressions, with a reproducible anomaly showing the ordered sequence of primes fills residue classes faster than any exchangeable resampling of itself; a finite- x variance law for prime counts in microscopic windows with second-order constant $\gamma + \log 2\pi - 1$ (a sharpening of Montgomery–Soundararajan into Gallagher’s regime, with numerical precedent); a candidate local-global law for the mixed polynomial–exponential sequence $n^2 + 2^n$; a liminf first-occurrence law for prime gaps with the Granville-corrected constant $\sqrt{e^\gamma/2} = 0.943682\dots$, registered against Shanks’s classical constant 1; and, replacing four statements the review judged bibliographically novel at most, the power-obstruction ladder for $n = p + j^k$ (total algebraic obstruction for composite k , a Bateman–Horn lane for prime k —the dichotomy itself corrected by a third external review, which refuted our first even/odd version), the finiteness of primorial twins (the convergent-side Borel–Cantelli exemplar), and a Goldbach lane race in which prime-square contamination pushes every bias term into the $(1, 1)$ lane.

*All constants, counts, and audit results in this paper are reproducible from the accompanying code repository: `run_all.py` regenerates every number. Primality of integers beyond 3.3×10^{24} was tested with Baillie–PSW; all smaller primality claims are deterministic (fixed-base Miller–Rabin).

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1 Introduction

A good conjecture in the theory of numbers is not merely a statement that has resisted counterexample. It is the *prediction of a model*: the assertion that the primes, beyond their local structure, behave like a random set of density $1/\log n$. The discipline this imposes is exacting. A conjecture worth stating should (i) survive every congruence and size obstruction, which is a computation, not a hope; (ii) come with a quantitative asymptotic whose constant is derived, not fitted; (iii) demand exactly the fluctuations the model earns—square-root noise with logarithmic corrections—and no fewer (the Mertens conjecture died of demanding fewer); (iv) sit inside the standard hierarchy of conjectures (Hardy–Littlewood k -tuples $\subset$ Schinzel’s Hypothesis H $\subset$ Bateman–Horn), so that its failure would propagate; and (v) be falsifiable instance-by-instance by computation. Numerical range is the least important column in the ledger: most deep phenomena drift at the rate $\log \log x$, and $\log \log 10^{18} \approx 3.7$, so verification “to 10^{18} ” is weak evidence by itself. What persuades is the *shape* of the agreement—a count that tracks a derived constant through several decades with residuals that look like noise.

The twenty-five statements of this paper were produced systematically to those specifications, together with the audit that the specifications demand. They are organized in four blocks: Bateman–Horn systems (§3), barely-divergent sparse sequences (§4), representation problems with Borel–Cantelli accounting (§5), and statistical laws of the prime sequence (§6). Section 7 describes the adversarial audit: what was attacked, what survived, and in one case what did not. We consider the audit trail part of the mathematical content. Its headline is instructive: the one outright refutation (Conjecture 13, first version) was produced not by bad luck but by a density-zero algebraic family—the cubes—invisible to a probability model that integrates over density. This is precisely the classical lesson of admissibility, relearned here in public.

Two further audit outcomes deserve emphasis. First, an independent literature layer found that fifteen of the twenty-five statements already exist in essence—some verbatim (Dubner’s twin-sum conjecture, the Stern list for $p + 2k^2$, Caldwell–Gallot’s $e^\gamma \log N$ laws), one with our exact constant catalogued as an OEIS entry (A188596 for Conjecture 6). We keep these statements in the paper, restated in our uniform framework and re-verified at our bounds, but they are attributed explicitly in §8; the residual claim to novelty of this paper rests on the other ten.

Second, several of our *first drafts* were corrected by data before any external audit: the naive Poisson claim behind Conjecture 20 fails at every accessible scale (the corrected second-order law is the conjecture); the first-guess bias direction in the twin race of Conjecture 21 was reversed by a mod-3 computation; the naive Exp(1) mean in Conjecture 22 is off by a factor that took three controls to localize. We report these corrections because the difference between a fitted claim and a derived claim is exactly what an audit is for.

Taxonomy: benchmarks versus research conjectures

An external review of an earlier draft made a criticism we accept in full: an unrecorded specialization of Bateman–Horn is analogous to evaluating a classical special function at a new argument—worthwhile as data, but not a new conjecture in the conceptual sense. We therefore classify the twenty-five statements explicitly. *Benchmarks* (Conjectures 1–3, 5–8, 12–18, 23, 24): statements previously known in essence, or that follow mechanically from Hardy–Littlewood/Bateman–Horn once admissibility is checked; their role here is calibration—each carries a derived constant that the data must and do hit. *Research conjectures* (Conjectures 4, 9–11, 19–22, 25): statements involving a mechanism, law, or dichotomy we could not find in the literature, stated with their dependence assumptions explicit. Four slots in this second group are wholesale replacements: after the external review judged the original occupants (a quintuplet constellation, an arbitrary shifted quadratic pair, the Caldwell–Gallot primordial benchmark, and a restatement of Lemke Oliver–Soundararajan) bibliographically novel at most, they were replaced by the power-obstruction ladder (4), the twin cyclotomic bases (5, still benchmark-grade but structurally motivated), primordial twins (10), and the Goldbach lane race (25); each replacement was novelty-checked and verified before admission. Within the research group we regard Conjecture 22(ii), Conjecture 21, and Conjecture 20 as the paper’s core, joined now by 25 (same mechanism family as 21) and 4; Conjectures 9, 11 and 19 carry open modeling questions flagged in place.

Verification standards

All counts below are exact (sieves; deterministic Miller–Rabin for $n < 3.3 \times 10^{24}$; Baillie–PSW beyond, flagged as such). Every comparison reports the Poisson-normalized residual $z = (\text{obs} - \text{pred})/\sqrt{\text{pred}}$. For sparse counts the cumulative comparison is contaminated at small x by the phantom mass of the main-term integral near its lower limit; in those cases we also report per-decade increments, which are clean. Constants are computed by truncated Euler products with the truncation wobble (difference between cutoffs) reported as an error bar; for conditionally convergent quadratic and cubic products this wobble, not the last digit, is the honest precision.

2 Notation

p, q denote primes; γ is Euler’s constant, $e^\gamma = 1.781072\dots$; φ is Euler’s totient; $\phi = (1 + \sqrt{5})/2$; F_n the Fibonacci numbers; $p\#$ the primorial. For irreducible $f_1, \dots, f_k \in \mathbb{Z}[x]$ with positive leading coefficients, the Bateman–Horn singular series is

$$C(f_1, \dots, f_k) = \prod_p \frac{1 - \omega(p)/p}{(1 - 1/p)^k}, \quad \omega(p) = \#\{n \bmod p : p \mid f_1(n) \cdots f_k(n)\},$$

and the corresponding main term is

$$I(N) = \int_2^N \frac{dt}{\prod_i \log f_i(t)},$$

which absorbs the usual $1/\prod_i \deg f_i$. The Bateman–Horn conjecture [2] asserts $\#\{n \leq N : \text{all } f_i(n) \text{ prime}\} \sim C \cdot I(N)$ whenever $C \neq 0$; every conjecture in §3 is an instance with its constant evaluated. For nonlinear systems the product converges only conditionally; throughout, it is taken in the canonical ordering of increasing primes, in which convergence follows from the equidistribution of $\omega(p)$ about its Chebotarev mean. Our numerical truncation wobble is an error bar for this canonically ordered product, not a substitute for that convergence statement. We write $\text{Li}_k(x) = \int_2^x (\log t)^{-k} dt$. The twin singular series is $\mathfrak{S}(d) = 2C_2 \prod_{p|d, p>2} \frac{p-1}{p-2}$ for even d , with $2C_2 = 1.3203236\dots$; the Goldbach series $\mathfrak{S}(n)$ is the same product over odd $p \mid n$.

Admissibility ($\omega(p) < p$ for all p) is asserted programmatically during every constant computation. The check does real work: the natural-looking pair $\{n^2 + n + 1, n^2 + n + 3\}$ has $\omega(3) = 3$ and is rejected; Conjecture 5 therefore uses the nearest admissible companion instead.

Novelty labels, from the independent literature audit (§8): (a) previously stated in essence; (b) classical family, quantified instance apparently unstated.

3 Bateman–Horn instances

The eight conjectures of this block share one hypothesis-generating move: choose an admissible system of polynomials, compute its singular series, and state the count. The heuristic is uniform, so we give it once. Model the primality of the values $f_i(n)$ as independent events of probability $1/\log f_i(n)$, corrected by the factor $(1 - \omega(p)/p)/(1 - 1/p)^k$ at each prime p to account for the joint local behaviour; multiplying the corrections and integrating gives $C \cdot I(N)$.

Conjecture 1 (Quadratic twin pair; (a), cf. OEIS A080149, [28]). $\#\{n \leq N : n^2 + 1, n^2 + 3 \text{ both prime}\} \sim C_1 I(N)$, $C_1 = C(x^2 + 1, x^2 + 3) = 2.954014\dots$

Here n must be even and $n \not\equiv 0 \pmod{3}$; for $p \geq 5$, $\omega(p) = 2 + (\frac{-1}{p}) + (\frac{-3}{p}) \leq 4 < p$. The conjecture implies infinitely many twin pairs $(m + 1, m + 3)$ with m a perfect square—a de Polignac refinement along a quadratic sequence.

Conjecture 2 (Cubic shift; (b)). $\#\{n \leq N : n^3 + 2 \text{ prime}\} \sim C_2^* I(N)$, $C_2^* = C(x^3 + 2) = 1.298435 \pm 0.0003$.

For $p \equiv 2 \pmod{3}$ the cube map is a bijection mod p , so $\omega(p) = 1$ *exactly* and the local factor is $(1 - 1/p)^{-1}(1 - 1/p) = 1$: only $p \equiv 1 \pmod{3}$ (where $\omega \in \{0, 3\}$ according as -2 is a cubic residue) moves the product, which therefore converges only conditionally—hence the quoted wobble. No statement of Bunyakovsky type is proven for any cubic; the nearest theorem is Heath-Brown’s $x^3 + 2y^3$ [3], which is the partial-result neighbour that makes the single-variable statement a research programme rather than a monolith.

Conjecture 3 (Chernick chain; (a), Dubner [8]). $\#\{p \leq x : p, 2p - 1, 3p - 2 \text{ all prime}\} \sim C_3 I(x)$, $C_3 = C(x, 2x - 1, 3x - 2) = 2.858249\dots$

The three primes form an arithmetic progression with common difference $p - 1$, first instance $(3, 5, 7)$; for $p = 6m + 1$ the product $p(2p - 1)(3p - 2) = (6m + 1)(12m + 1)(18m + 1)$ is Chernick’s universal-form Carmichael number, so this conjecture quantifies the population from which the classical Carmichael constructions draw.

Conjecture 4 (The power-obstruction ladder, corrected; apparently new as a family). *For $k \geq 2$ and $m \geq 2$, write $D_k(m) = m^k - (m-1)^k$, and call m^k representable if $m^k = p + j^k$ for some prime p and $j \geq 1$. Then: (i) [theorem] for composite k , no k -th power is representable: if r is a proper divisor of k , then for every $1 \leq j < m$ the difference $m^k - j^k$ has the proper factor $m^r - j^r > 1$ with cofactor > 1 ; (ii) [elementary] for prime k (including $k = 2$), m^k is representable if and only if $D_k(m)$ is prime; and D_k is irreducible for prime k (its roots $\zeta^j/(\zeta^j - 1)$, ζ a primitive k -th root of unity, form a single Galois orbit— D_k is a linear-fractional transform of the cyclotomic polynomial Φ_k); (iii) [conjecture] for each prime k the representable lane follows Bateman–Horn: $\#\{m \leq M : D_k(m) \text{ prime}\} \sim C_k \int_2^M dt / \log D_k(t)$.*

Finding 1. Our first version of this ladder claimed an even/odd dichotomy, with a Bateman–Horn lane for every odd k . The third external review refuted it: for odd composite $k = rs$ the factorization in (i) applies (e.g. $D_9(m)$ is *never* prime), so the lane exists only for prime k . We verified the refutation directly ($k = 9, 15, 21, 25, 27, 33$: zero prime values of D_k to $m = 2000$, with the predicted divisor $D_r(m) \mid D_k(m)$ checked identically) and record that this is the second algebraic-factorization failure the audit programme has caught in our own output—the exact failure mode our procedure was built to detect, twice missed at generation time and twice caught downstream.

The family grew out of the audit refutation behind Conjecture 13 (the case $k = 3$). It replaces, at the suggestion of an external review, a bibliographically-novel-at-most quintuplet constellation that formerly occupied this slot (its data survive in the repository; its $4 \cdot 10^9$ verification, ratio 1.0004, remains part of the audit record as Hardy–Littlewood calibration).

Conjecture 5 (Twin cyclotomic bases; no prior trace found). *There are infinitely many n for which the length-3 repunits in bases n and $n + 1$ are simultaneously prime: $\#\{n \leq N : \Phi_3(n) = n^2 + n + 1 \text{ and } \Phi_3(n+1) = n^2 + 3n + 3 \text{ both prime}\} \sim C_5 I(N)$, with $C_5 = C(x^2 + x + 1, x^2 + 3x + 3)$.*

Both forms are odd for every n (each is Φ_3 of an integer), and mod 3 each vanishes on exactly one class ($n \equiv 1$ and $n \equiv 0$ respectively), so the pair is admissible with $\omega(3) = 2$. Unlike the arbitrary shifted pair that formerly occupied this slot (replaced at the suggestion of an external review), the companion here is structurally forced: it is the same cyclotomic value at the consecutive base, making this the natural “twin” statement inside the repunit family of Conjecture 6.

Conjecture 6 (Cyclotomic Sophie Germain; (a); the constant is OEIS A188596). $\#\{p \leq x : p^2 + p + 1 \text{ prime}\} \sim C_6 I(x)$, $C_6 = C(x, x^2 + x + 1) = 1.521661 \pm 0.0005$ (catalogue value 1.5217315..., OEIS A188596).

Since $p^2 + p + 1 = (p^3 - 1)/(p - 1)$ is the length-3 repunit in base p , the conjecture says: infinitely many prime bases have prime repunits of length 3. This is the base-independent home of the phenomenon whose base-2 shadow is the Mersenne sequence. Locally, $3 \mid p^2 + p + 1$ exactly when $p \equiv 1 \pmod{3}$, so half of all primes are barred at 3; for $q \equiv 1 \pmod{3}$ the two primitive cube roots of unity give $\omega(q) = 3$ (including the root from the linear factor).

Conjecture 7 (Downward companion; (b); sequence is OEIS A062326). $\#\{p \leq x : p^2 - 2 \text{ prime}\} \sim C_7 I(x)$, $C_7 = C(x, x^2 - 2) = 3.383216 \dots$

Squares are $\equiv 0, 1 \pmod{3}$, so $3 \nmid p^2 - 2$ always—no loss at 3—and $\omega(q) = 1 + (\frac{2}{q})$ for the quadratic factor. The same pairs $(p, p^2 - 2)$ reappear as the deterministic term of the

twin race in Conjecture 21: this conjecture’s counting function sets that term’s magnitude. (The dependence is one-directional and partial—a race bias could conceivably be established in averaged form without the full asymptotic here—but a refutation of this conjecture would remove Conjecture 21’s predicted mechanism.)

Conjecture 8 (Quadratic triple; (b); singular series possibly new). $\#\{n \leq N : n^2 + 1, n^2 + 3, n^2 + 7 \text{ all prime}\} \sim C_8 I(N)$, $C_8 = 10.64599 \pm 0.006$.

This refines Conjecture 1 exactly as quadruplets refine twin primes, one level up from the linear world. Mod 7: $n^2 + 7$ bars $n \equiv 0$, $n^2 + 3$ bars $n \equiv \pm 2$, and $n^2 + 1$ has no roots, so $\omega(7) = 3$. The extension by $n^2 + 9$ remains admissible (its singular series absorbs the resulting confinement to $n \equiv 0 \pmod{5}$); indeed the third external review reports that the longer simultaneous pattern $n^2 + \{1, 3, 7, 9, 13\}$ has been recorded in the OEIS since 2000, so the tuple *family* has prior presence and only the singular-series evaluation here is plausibly new—our earlier “no prior trace” claim was too strong and is withdrawn.

Verification

	bound	observed	predicted	ratio	z
C1	10^7	32,898	32,862.7	1.0011	+0.19
C2	10^7	287,956	287,784.8	1.0006	+0.32
C3	$3 \cdot 10^8$	125,379	125,429.4	0.9996	−0.14
C4 _{k=2}	10^7	1,270,606	1,270,902.8	0.9998	−0.26
C4 _{k=3}	10^6	126,826	126,641.6	1.0015	+0.52
C4 _{k=5}	$8 \cdot 10^5$	56,925	57,021.3	0.9983	−0.40
C5	10^7	33,274	32,976.7	1.0090	+1.64
C6	10^7	33,661	33,855.5	0.9943	−1.06
C7	10^7	74,914	75,273.8	0.9952	−1.31
C8	10^7	3,963	3,998.6	0.9911	−0.56

For C4: the even-rung theorem was checked numerically to 10^8 (no fourth power is $p + j^4$); the three verifiable Bateman–Horn lanes $k = 2, 3, 5$ (constants 2 exactly, 3.36181, and 3.67770 ± 0.007 , the quartic product computed by brute-force root counts to 10^5) all sit within $|z| < 0.6$. For C5, $C_5 = 2.964239 \pm 0.002$. Independent clean-context recomputation gave $C_6, C_7, C_8 = 1.521730, 3.383227, 10.647706$ with primes to 10^9 (each inside our stated wobble) and independent counts to 3×10^7 with final ratios 0.9962, 0.9965, 1.0059. (The quintuplet that formerly occupied slot 4 retains its audit-record calibration: observed 15,236 against predicted 15,230.3 at $4 \cdot 10^9$, ratio 1.0004, $z = +0.05$.)

4 Sparse sequences: barely-divergent Borel–Cantelli

For sequences whose n th term has size e^{cn} the expected number of primes is $\sum_n \kappa_n / (cn)$: convergence or divergence of this sum is the entire question, and when it diverges it diverges logarithmically, so the right conjecture has counting function $\alpha \log x$ with a derived α —never more.

Conjecture 9 (Fibonacci primes; cf. [5]). $\#\{p \leq x : F_p \text{ prime}\} \sim c_F \log x$ for a constant $c_F > 0$ determined by the local (rank-of-aparition) structure of the Fibonacci sequence in the

sense of the Grantham–Granville recurrence-sequence model. The first-order screening value is $c_F^{(0)} = e^\gamma / \log \phi = 3.70122\dots$; the data below indicate $c_F < c_F^{(0)}$, and we deliberately do not conjecture the exact constant.

Every prime factor of F_p ($p \neq 5$ prime) has rank of apparition exactly p , hence is $\equiv \pm 1 \pmod{p}$: the Lenstra–Pomerance–Wagstaff screening [6] transfers with $\log 2 \mapsto \log \phi$. The data sit persistently *below* the constant: 25 Fibonacci prime indices $\leq 10^4$ (independently reconfirmed by a second implementation) against a predicted 34.1, ratio ≈ 0.73 , $z = -1.6$, stable over three decades. Grantham and Granville [5] caution that naive constants for such sequences require $O(1)$ local corrections; the mod-5 divisor structure of F_p is the natural suspect, and we flag this constant as the component of the suite most likely to need a Granville-type patch.

Conjecture 10 (Primorial twins; apparently new). *Only finitely many primes p have $p\# - 1$ and $p\# + 1$ simultaneously prime, and the complete list is $p \in \{3, 5, 11\}$ (i.e. the primorials 30 and 2310 are, with 6, the only primorials lying between twin primes).*

Each event separately has Caldwell–Gallot probability $\sim e^\gamma \log p/p$ [4] and divergent sum (the classical benchmark that formerly occupied this slot, retained inside Conjecture 12’s family); the *joint* event has probability $\sim (e^\gamma \log p/p)^2$, whose sum over primes converges. This is the suite’s exemplar of the convergent side of the Borel–Cantelli dichotomy—the accounting that predicts finitude rather than infinitude—and we could find no prior statement of it. The model-expected number of examples beyond $p = 4000$ is $\approx 7 \times 10^{-3}$ (computed tail). The defensible core is finiteness; the exact three-element list is the Goldilocks-maximal form—strictly stronger than the tail estimate can guarantee, and chosen deliberately as the most falsifiable statement.

Conjecture 11 ($n^2 + 2^n$; sequence is OEIS A064539). *Every prime of the form $n^2 + 2^n$ with $n > 1$ has $n \equiv 3 \pmod{6}$ (elementary), and: (i) infinitely many such primes exist, with counting function $\asymp \log N$; (ii) [candidate law] the count is $\sim \sum_{n \leq N, n \equiv 3 \pmod{6}} \kappa / \log(n^2 + 2^n)$, where $\kappa = \prod_p (1 - \delta_p) / (1 - 1/p)$, δ_p the exact density of $n \equiv 3 \pmod{6}$ with $p \mid n^2 + 2^n$ over the period $\text{lcm}(6, p \cdot \text{ord}_p 2)$, and—this is part of the conjecture—the product converges in increasing-prime order. Truncation at $p \leq 300$ gives the candidate value $\kappa = 4.2734\dots$*

This is not a Bateman–Horn problem: the local conditions at different primes are tied through the orders $\text{ord}_p 2$ and need not behave independently, and no standard convergence theory applies to the κ -product. We therefore separate the robust claim (i) from the candidate law (ii), and note that the eight hits below 4200 are far too few to validate a four-digit constant—the verification below checks consistency, not the constant.

The congruence claim is elementary: n odd is forced by parity, and for odd n with $3 \nmid n$ we have $n^2 + 2^n \equiv 1 + 2 \equiv 0 \pmod{3}$. What appears to be a harsh obstruction is, inside the surviving lane, a local bonus: $\delta_2 = \delta_3 = 0$ contribute the factors $2 \cdot \frac{3}{2} = 3$ to κ . Mixed polynomial–exponential sequences are exactly where hand-waving fails and the computed local product earns its keep.

Conjecture 12 (Factorial primes; (a), [4]). $\#\{n \leq N : n! + 1 \text{ prime}\} \sim e^\gamma \log N$.

Verification

C9: hits $\{3, 5, 7, \dots, 9311, 9677\}$, 25 of predicted 34.1 ($z = -1.56$; see the remark above). C10: exhaustive to $p \leq 4000$: the twins are exactly $\{3, 5, 11\}$; individually, $p\# + 1$ is prime for

$p \in \{2, 3, 5, 7, 11, 31, 379, 1019, 1021, 2657, 3229\}$ (11 against the Caldwell–Gallot prediction 14.8, $z = -0.98$) and $p\# - 1$ for $p \in \{3, 5, 11, 13, 41, 89, \dots\}$, and the two divergent single-sided laws straddle their shared constant while the joint count stops at three—the convergent accounting in action. C11: hits $\{3, 9, 15, 21, 33, 2007, 2127, 3759\}$ up to 4200: observed 8, model 8.19, $z = -0.07$; the lane constraint was verified exhaustively (no prime off $n \equiv 3 \pmod{6}$ to $n = 3000$, independently reconfirmed). C12: 15 primes with $n \leq 700$ against 11.7 ($z = +0.98$), completing the $e^\gamma \log$ family benchmark alongside the single-sided primorial counts above.

5 Representation problems

Theorem 1 (Cube obstruction; found by the adversarial audit). *For $k \geq 2$, the cube k^3 is representable as $p + j^3$ with p prime, $j \geq 1$, if and only if $3k^2 - 3k + 1$ is prime. Consequently the set of integers not representable as $p + k^3$ is infinite: the non-representable cubes alone have counting function $\sim x^{1/3}$. (That the full exceptional set is $\sim x^{1/3}$ follows only when combined with Conjecture 13(i), and is conjectural.)*

Proof. $k^3 - j^3 = (k - j)(k^2 + kj + j^2)$; for $1 \leq j \leq k - 2$ both factors exceed 1, so the difference is composite. The only candidate is $j = k - 1$, giving $3k^2 - 3k + 1$. Since $3k^2 - 3k + 1$ is composite for a density-one set of k (all but $O(K/\log K)$ of $k \leq K$, by Chebyshev bounds for the quadratic’s prime values), all but a vanishing proportion of cubes are unrepresentable. $\square$

Finding 2. As first generated (“the exception set of $n = p + k^3$, $k \geq 1$, is finite”), the next conjecture was **false**, and our own adversarial battery refuted it: all 412 exceptions found in $(10^8, 10^9]$ are perfect cubes, explained by Theorem 1. The error is a textbook violation of the admissibility clause of the design criteria—an algebraic obstruction along a density-zero subsequence, invisible to a Borel–Cantelli sum taken over all n . We record the failure rather than hide it; the restated version separates theorem from conjecture.

Conjecture 13 (Prime plus a cube, restated; (a)-core: the non-cube finiteness is Hardy–Littlewood’s $E_3(X) = O(1)$ [1]). (i) *The set of non-cube integers $n \geq 2$ not representable as $p + k^3$ ($k \geq 1$) is finite.* (ii) *The representable cubes obey Bateman–Horn: $\#\{k \leq K : 3k^2 - 3k + 1 \text{ prime}\} \sim C(3x^2 - 3x + 1)I(K)$.*

(Empirically the non-cube exception census by decade is 4, 27, 168, 763, 2011, 2808, 1181, 88 up to 10^8 , peaking at the sixth decade and collapsing thereafter, with largest element found 78,526,384; these are verification data, not part of the formal statement.)

Conjecture 14 (Stern list; (a): OEIS A060003). *Exactly ten odd integers are not of the form $p + 2k^2$ with $k \geq 1$:*

$$1, 3, 17, 137, 227, 977, 1187, 1493, 5777, 5993.$$

The expected number of representations of odd n is $\asymp \sqrt{n}/\log n$, so the expected number of exceptions beyond any X decays like $\sum_{n>X} \exp(-c\sqrt{n}/\log n)$: the list should be complete, and the seven prime members are precisely the Stern primes. Verified here to 10^9 , one decade past our original bound, with no eleventh exception.

Conjecture 15 (Goldbach in the $(3, 3)$ lane; (a), K. Martin [9]). *Every $n \equiv 2 \pmod{4}$ with $n \geq 6$ is $p + q$ with $p \equiv q \equiv 3 \pmod{4}$, and the ordered representation count obeys*

$$R_3(n) \sim \frac{1}{2} \mathfrak{S}(n) \int_3^{n-3} \frac{dt}{\log t \log(n-t)}.$$

Normalization: $R_3(n)$ counts *ordered* pairs (p, q) , and the full-interval integral is the ordered main term for all Goldbach representations, so the factor $\frac{1}{2}$ is exactly the equidistribution split between the (1, 1) and (3, 3) lanes (parity forces $p \equiv q \pmod{4}$ when $n \equiv 2 \pmod{4}$). The (3, 3) lane keeps its smallest representations $(3 + q)$ and is the lane with *no* exceptions at all, which is why the uniform threshold $n \geq 6$ can be conjectured. Verified exhaustively to 10^9 ; the count formula matches to 0.2–1.3% at $n \approx 10^6, 10^7, 10^8$.

Conjecture 16 (Uniform quantitative de Polignac; (a)-core). (i) $\pi_d(x) = \#\{p \leq x - d : p, p + d \text{ prime}\} = \mathfrak{S}(d) \text{Li}_2(x) (1 + o(1))$ uniformly for even $d \leq (\log x)^A$. (ii) [residual law] With $z_d(x) = (\pi_d(x) - \mathfrak{S}(d) \text{Li}_2(x)) / \sqrt{\mathfrak{S}(d) \text{Li}_2(x)}$ and $D(x) = (\log x)^A$, the empirical measure $D(x)^{-1} \sum_{2|d \leq 2D(x)} \delta_{z_d(x)}$ converges weakly, as $x \rightarrow \infty$, to a centered Gaussian whose variance $\sigma^2 \leq 1$ is determined by the covariance structure of overlapping pair counts (the same primes participate in many π_d ; Poisson normalization therefore need not give unit variance), with no trend in d . Empirically $\sigma \approx 0.5$ at $x = 10^8$.

The force of the statement is its uniformity: the singular series takes a thousand different values over $d \leq 2000$, spanning a factor of about three between $d = 2$ and the primorial-rich d , and the counts must reproduce the entire profile simultaneously with square-root errors. A conspiracy would need to bend a thousand-dimensional vector, not one number. At $x = 10^8$: correlation 0.999999, slope 0.99966, $\max_d |z_d| = 1.80$; stressed to $d \leq 6000$, $\max |z_d| = 2.01$ over three thousand values.

Conjecture 17 (Dubner’s conjecture; (a) [7]). *Every even $n \geq 4210$ is a sum of two members of twin-prime pairs; the 35 exceptions (OEIS A007534, largest 4208) are complete.*

Twin members have density $\asymp 1/\log^2 x$, so expected representations grow like $n/\log^4 n$ and exceptions must be finite; the observed exceptions cluster in triples of consecutive even numbers, a structural fingerprint of the sparse low-lying twin population. Verified to 10^8 , adversarially re-swept to 10^9 , and independently re-verified to 10^9 by a second implementation with a different algorithm: the same 35 exceptions each time.

6 Statistical laws

Conjecture 18 (Twin-gap records; (a), Kourbatov [11]). (i) *The maximal gap $G_t(x)$ between consecutive twin-prime starts up to x satisfies $G_t(x) \asymp \log^3 x$.* (ii) [candidate constant] $\limsup_x G_t(x)/\log^3 x = 1/(2C_2) = 0.757390\dots$ Part (i) is the robust claim; the exact constant in (ii) inherits the known fragility of Cramér-style extreme-value constants (Granville’s correction arose precisely because extreme events feel local sieve scales invisible to an inhomogeneous Poisson model), and is offered as a working value, not an assertion at the same confidence as (i).

Twin starts thin the integers to density $2C_2/\log^2 t$; a Cramér process at that density has record waiting times $\sim (\text{mean spacing}) \times \log(\text{number of events}) = \log^3 x / 2C_2$. The Cramér–Granville episode teaches that such constants can be shifted by local corrections (Granville’s $2e^{-\gamma}$ [13]); the claim to take seriously is the cube of the logarithm, and the constant is flagged as first-order. Records to $4 \cdot 10^9$ (largest observed twin-to-twin gap 5292, following the twin at 2,466,641,069) wander in $[0.41, 0.56] \cdot \log^3 x$, approaching the asymptote from below on a log log clock, as such statistics must.

Conjecture 19 (First occurrences of prime gaps, liminf form; (b)). *Let $p(g)$ be the prime starting the first occurrence of a prime gap of exactly g . Then: (i) $\log p(g)/\sqrt{g}$ is bounded between positive constants; (ii) [liminf law]*

$$\liminf_{g \rightarrow \infty} \frac{\log p(g)}{\sqrt{g}} = \sqrt{e^\gamma/2} = 0.943682\dots,$$

the liminf being attained along gap sizes realized at the Cramér–Granville maximal-gap envelope.

An earlier draft asserted the full limit, inverting Granville’s $\limsup G(x)/\log^2 x = 2e^{-\gamma}$; an external review correctly observed that this inversion is invalid. A limsup envelope for maximal gaps controls only where gaps *can first appear at the earliest*, i.e. a lower envelope for $p(g)$ —hence a liminf claim, not a limit. Whether $\log p(g)/\sqrt{g}$ converges at all, and if so whether the limit is Shanks’s classical 1 [12] (as Wolf’s refinement $p(g) \sim \sqrt{g}e^{\sqrt{g}}$ and the Kourbatov–Wolf residue-class laws [21] also assert) or the envelope value, involves at least three distinct phenomena—whether a given g occurs near the extreme scale at all, the distribution of exact gap lengths near records, and the difference between record gaps and non-record first occurrences—and is left open here. The two candidate constants differ by 6%, beneath the resolution of any feasible computation: our regression slope over $g \geq 100$ falls from 1.307 at 10^9 to 1.297 at $4 \cdot 10^9$, far above both and decreasing, as it must be for either.

Conjecture 20 (Microscopic variance law; (b); the constant is Montgomery–Soundararajan’s [15]). *Fix $\lambda > 0$ and $c \in (0, 1]$, and let X count primes in $[t, t + \lambda \log x]$ for t uniform on $[x, (1+c)x]$. Then, under the Hardy–Littlewood k -tuple conjectures, $X \Rightarrow \text{Poisson}(\lambda)$ (Gallagher’s argument [14], which is conditional on exactly those conjectures), and at finite x*

$$\frac{\text{Var } X}{\mathbb{E} X} = 1 - \frac{\log(\lambda \log x) + \gamma + \log 2\pi - 1}{\log x} + o\left(\frac{1}{\log x}\right),$$

with $\gamma + \log 2\pi - 1 = 1.41509\dots$, the leading correction being independent of the sampling constant c .

The deficit arises by inserting Montgomery’s singular-series average $\sum_{d \leq h} \mathfrak{S}(d)(h-d) = \frac{h^2}{2} - \frac{h}{2}(\log h + \gamma + \log 2\pi - 1) + o(h)$ into the second factorial moment at the microscopic window $h = \lambda \log x$. Three cautions, prompted by external review, are built into the statement: Gallagher’s Poisson law is conditional, not a theorem about primes; the sampling measure for t is specified, since different weightings shift lower-order terms; and the error is claimed only as $o(1/\log x)$ —upgrading it to $O(1/\log^2 x)$ would require a quantitative remainder in the singular-series average together with control of prime powers, endpoint effects, and the variation of $\log t$ across the sampling range, none of which the $o(h)$ form supplies. The naive claim $\text{Var}/\mathbb{E} = 1$ fails at every accessible scale by 18–28%; the corrected law matches with no fitted parameter, and the observed residual $\approx +0.003$ is consistent with (but not evidence for) a genuine $1/\log^2 x$ term. (Numerical precedent: Sanchis-Lozano [22] fitted the Montgomery–Soundararajan variance shape at $x \sim 10^8$ in the mesoscopic regime $h \gg \log N$; the microscopic statement and test appear to be new, and we claim no more than that.)

λ	$x = 10^9$		$x = 4 \cdot 10^9$	
	observed	predicted	observed	predicted
1/2	0.8241	0.8189	—	—
1	0.7910	0.7854	0.7998	0.7960
2	0.7527	0.7520	0.7670	0.7646
4	0.7201	0.7185	—	—

The residual $\approx +0.003$ is the anticipated $O(1/\log^2 x)$ term.

Conjecture 21 (Twin race modulo 5; (b), apparently new). *Twin starts $p > 5$ lie in classes $p \equiv 1, 2, 4 \pmod{5}$. Since $3 \mid q^2 + 2$ for every prime $q > 3$, the only pairs $(n, n+2)$ with one entry a prime square that contribute prime-power terms to the weighted count $\sum_{n \leq x} \Lambda(n) \Lambda(n+2)$ are $(q^2 - 2, q^2)$; and $q^2 - 2 \pmod{5} \in \{2, 4\}$, never 1. Consequently, with $D_1(x) = \pi_t(x; 5, 1) - \frac{1}{2}(\pi_t(x; 5, 2) + \pi_t(x; 5, 4))$: (i) class 1 carries a systematic surplus*

$$D_1^{\text{sys}}(x) = \frac{1}{2 \log^2 x} \sum_{\substack{q \leq \sqrt{x} \\ q^2 - 2 \text{ prime}}} \log q \log(q^2 - 2),$$

whose scale is governed by the constant C_7 of Conjecture 7; (ii) classes 2 and 4 carry no deterministic term: $\pi_t(x; 5, 2) - \pi_t(x; 5, 4) = o(\sqrt{x}/\log^2 x)$ in logarithmic mean; (iii) the deterministic term is smaller than the random-walk noise $\sqrt{\pi_t(x)}$ by a factor $\log x$; consequently—in contrast to Chebyshev’s races—the twin race has no persistent leader: if the logarithmic density of $\{D_1 > 0\}$ exists, it equals $\frac{1}{2}$, with a positive finite- x excess of order $1/\log x$ from the drift-to-noise ratio; (iv) [mod-8 companion] since $q^2 \equiv 1 \pmod{8}$ for every odd q , the same mechanism puts the entire deterministic term on the single class 7 (mod 8): with $D_7(x) = \frac{1}{3} \sum_{a \in \{1, 3, 5\}} \pi_t(x; 8, a) - \pi_t(x; 8, 7)$,

$$D_7(x) = \frac{1}{\log^2 x} \sum_{\substack{q \leq \sqrt{x} \\ q^2 - 2 \text{ prime}}} \log q \log(q^2 - 2) + O^*(\sqrt{\pi_t(x)})$$

(twice the mod-5 term, undiluted), while classes 1, 3, 5 are mutually symmetric. The mod-5 and mod-8 races are two projections of one mechanism, so their joint behaviour is a family test in the sense of the shape-of-agreement criterion.

An earlier draft of clause (iii) asserted a limiting sign density strictly between $\frac{1}{2}$ and 1; an external review correctly observed that this is inconsistent with the scale analysis—when drift/noise $\rightarrow 0$, a Gaussian model forces the sign density to $\frac{1}{2}$, and a nontrivial limiting density would require a persistent normalized bias that no mechanism here produces. The corrected clause is what the model actually supports. Three further caveats are open modeling work: the passage from the Λ -weighted count to unweighted twin counts needs a careful derivation; higher prime powers contribute only at $x^{1/3}$ and below (cubes $q^3 \pm 2$), so no other configuration enters at the $\sqrt{x}$ scale, but this should be proved within the model; and a full race statement requires the covariance of the three class counts, which we have not modeled.

The mechanism is the twin analogue of the classical explanation of Chebyshev’s bias: prime-square terms contaminate the classes that can contain them. (Biases of twin primes in residue classes have been studied before—Sahoo [23] for the mod-4 race, and [24] for prime pairs versus isolated primes—but we could find no prior statement of the mod-5 race, the $(q^2 - 2, q^2)$ mechanism, or the quantified surplus below.) Mod 3 kills the pattern $(q^2, q^2 + 2)$ outright, an amusing local accident that halves the mechanism and (as a mod-3 computation showed, overturning a first guess of a quadratic-residue bias against class 4) concentrates the surplus entirely on class 1. Data to $4 \cdot 10^9$: class counts (3980017, 3982505, 3981922), D_1 within one noise unit of D_1^{sys} throughout, no persistent leader, and the control race $\pi_{t,2} - \pi_{t,4}$ wandering like a fair coin. Mod-8 data to 10^9 : class counts (856684, 855807, 856046, 855967) for $a = 1, 3, 5, 7$; $D_7 = +212$ at 10^9 against predicted $T = +254$ with noise ~ 1068 , positive through most of the range (log-density 0.775), and the three pairwise $\{1, 3, 5\}$ controls all within one noise unit of zero—the two projections of the mechanism behave coherently, none of it provable at present.

Conjecture 22 (Least primes in progressions; (a) for part (i) [16, 17, 26, 25], (b) for part (ii)). *Let $p(a, q)$ be the least prime $\equiv a \pmod{q}$ and $U(a, q) = \text{Li}(p(a, q))/\varphi(q)$. Then: (i) $U \Rightarrow \text{Exp}(1)$ marginally as $q \rightarrow \infty$; and, under the additional hypothesis that the class processes are asymptotically jointly independent in the extreme-value sense (which marginal convergence alone does not supply), $\max_a U - H_{\varphi(q)}$ converges to a Gumbel law ($H_n = \sum_{k \leq n} 1/k$), the order-statistics form of Wagstaff’s $\max_a p(a, q) \sim \varphi(q) \log^2 q$; (ii) the finite- q deficit, averaged over all reduced classes a at fixed q , decomposes at first order $1/\log q$ as*

$$1 - \mathbb{E}_a[U] = \frac{\theta_{\text{disc}}(q) + \theta_{\text{corr}}(q)}{\log q} + o\left(\frac{1}{\log q}\right),$$

where θ_{disc} is the discrete-hazard term (present already for Cramér pseudo-primes) and $\theta_{\text{corr}} > 0$ is the ordering term, which vanishes under any exchangeable reshuffling of the actual prime residues.

What is and is not specified: whether $\theta_{\text{disc}}, \theta_{\text{corr}}$ tend to constants, depend on the factorization of q (prime, squarefree, and highly composite moduli may differ), or vary slowly is open, and data at $q \leq 6000$ cannot distinguish $1/\log q$ from $(\log \log q)/\log q$ -type refinements—the measured aggregate $\theta \approx 1.5\text{--}1.9$ over $q \in [10^2, 6 \times 10^3]$, with $\mathbb{E}_a[U]$ bottoming near 0.737 around $q \approx 200$ and recovering through 0.776 by $q \approx 6000$, is calibration, not a determination of the functional form. A necessary comparison, raised by the third external review, is with Leung [25], who derives the exponential law for least primes under a uniform Hardy–Littlewood hypothesis and reports discrepancies from it for smooth moduli: whether θ_{corr} is a genuinely distinct ordering effect or a redescription of the singular-series corrections implicit in that analysis is an open question that part (ii)’s novelty claim depends on, and we flag it rather than assert it.

Remark 1. *The deficit is reproducible and is not an artifact of the time-change: a Cramér pseudo-prime control (independent Bernoulli “primes” with the correct densities and parity) shows roughly half of it—the discreteness of large per-candidate hazards—while the other half vanishes under any reshuffling of the actual prime residues (iid labels, or a random permutation of the true residue sequence, both restore $\mathbb{E}[U] = 1.00$). The ordered sequence of primes fills residue classes measurably faster than any exchangeable model. We measure θ ; we cannot yet derive it. It is the deepest unexplained number produced by this exercise.*

Conjecture 23 (Fermat quotients and the Wieferich ledger; (a), cf. [18]). *With $q_p = (2^{p-1} - 1)/p \pmod{p}$, three logically independent clauses: (i) [global equidistribution] q_p/p equidistributes on $[0, 1)$ with square-root discrepancy (Kolmogorov–Smirnov distance $O(\pi(x)^{-1/2})$, no drift); (ii) [shrinking targets] uniformly for $1 \leq K \leq p$, $\#\{p \leq x : q_p < K\} \sim \sum_{p \leq x} \min(1, K/p)$ —the case of fixed K , down to $K = 1$, is what governs the zero event and does NOT follow from (i), since $\{q_p = 0\}$ is a target of shrinking measure $1/p$; (iii) [Wieferich count, fluctuation-calibrated] $W(x) = \#\{\text{Wieferich } p \leq x\} = \log \log x + O_\varepsilon((\log \log x)^{1/2+\varepsilon})$, the fluctuation scale being that of the independent-Bernoulli model (variance $\sim \log \log x$).*

Two corrections from external review are absorbed here. First, an earlier draft derived (iii) from (i) with “hence”; that inference is invalid—a square-root-size error in the aggregate distribution is astronomically larger than $\log \log x$, and only the shrinking-target clause (ii) at bounded K speaks to the zero event. Second, the earlier fluctuation claim $O(1)$ violated this paper’s own calibration principle (criterion of demanding exactly the fluctuations the model earns): with event probabilities $1/p$ the count has standard deviation $\sim \sqrt{\log \log x}$, and the

statement now says so. We also retract any location prediction for a third Wieferich prime: the expected count reaching 3 “at doubly exponential heights” is an expected-count statement whose additive constant is uncalibrated, not a forecast. Consistency data: $\{1093, 3511\}$ below 10^8 against expected ≈ 2.7 ; $\sqrt{n}$ KS = 0.82, 0.82, 0.52, 0.63 at $x = 10^5, \dots, 10^8$ (bounded, no drift); census at $K = 100$: observed 169 against model 161.2.

Conjecture 24 (Conjecture F as a family; (a), [1], cf. [19]). *For odd A let $Q_A(N) = \#\{n \leq N : n^2 + n + A \text{ prime}\}$ and $C(A) = C(x^2 + x + A)$. Then, for every fixed $B > 0$, $Q_A(N) = C(A) I_A(N) (1 + o(1))$ uniformly over odd $1 \leq A \leq (\log N)^B$, with residuals obeying the analogue of Conjecture 16(ii): the empirical measure of $z_A(N) = (Q_A - C(A)I_A)/\sqrt{C(A)I_A}$ over the range converges to a centered Gaussian whose variance is set by the joint covariance of the family counts (sharing one set of primes correlates the Q_A ; a derived covariance kernel is required development, as the external review notes, and is not supplied here). (Uniformity over a range growing with N is the substantive claim; as an external review noted, uniformity over any fixed finite set of A is logically automatic from the individual asymptotics.) Among odd $A \leq 199$, Euler’s $A = 41$ has the largest constant, $C(41) = 6.6395463\dots$ (Cohen’s high-precision value [27]).*

At $N = 10^6$: mean ratio 1.0006, standard deviation 0.0027, $\max |z| = 1.92$, rank correlation 0.99988; for $A = 41$, observed 261,080 against predicted 261,017.6 ($z = +0.12$, using Cohen’s constant). A cautionary note from the referee audit belongs here: our own truncated Euler product for $C(41)$ at cutoff $2 \cdot 10^5$ reads 6.64092—wrong in the fourth significant decimal—because these conditionally convergent products drift at the 10^{-3} scale over practical cutoffs. Quoted digits for all quadratic-family constants in this paper are limited by the stated truncation wobble, not by the last printed digit. The entire 10^8 -bit primality dataset compresses to one product formula per A —the description-length virtue in its purest form.

Conjecture 25 (The Goldbach lane race; apparently new). *For $n \equiv 2 \pmod{4}$ let $R_1(n)$, $R_3(n)$ be the ordered Goldbach representation counts in the lanes $p \equiv q \equiv 1 \pmod{4}$ and $p \equiv q \equiv 3 \pmod{4}$ (Conjecture 15), and $D(n) = R_3(n) - R_1(n)$. Prime-square terms $n = q^2 + p$ (q odd prime) land exclusively in the (1,1) lane, since $q^2 \equiv 1 \pmod{4}$ and then $n - q^2 \equiv 1 \pmod{4}$ automatically. Consequently the (3,3) lane leads on average: with*

$$D_{\text{sys}}(n) = \frac{2}{\bar{\ell}(n)} \sum_{\substack{q \text{ odd prime, } q \leq \sqrt{n} \\ n - q^2 \text{ prime}}} \log q \log(n - q^2), \quad \bar{\ell}(n) = \frac{n - 6}{\int_3^{n-3} \frac{dt}{\log t \log(n - t)}},$$

(the analytic mean of $\log p \log(n - p)$ under the lane profile; the empirical lane mean is its estimator and is what the verification uses), then for n drawn logarithmically at random from $\{n \equiv 2 \pmod{4}, n \leq x\}$, $\mathbb{E}[D - D_{\text{sys}}] = o(\mathbb{E}[D_{\text{sys}}])$ as $x \rightarrow \infty$; per- n , D is D_{sys} plus noise of the square-root scale, so the sign of D favours the (3,3) lane only weakly at any fixed n .

Derivation status: the identity behind D_{sys} is the Λ -weighted lane symmetry minus its prime-power part; the transfer to unweighted counts is a partial-summation argument whose error terms (higher prime powers, which enter only at the $n^{1/3}$ scale, and the variation of the weight across the lane) remain to be written out—the same open transfer flagged at Conjecture 21, and the review is right that the conjecture’s maturity awaits it.

This is the Goldbach-partition analogue of Chebyshev’s bias, produced by the same explicit-formula mechanism as Conjecture 21 but through $q^2 + p$ patterns rather than twin patterns; the single-sign prediction (every square contamination falls into one lane) makes it cleaner than the

classical race. We could find no prior statement. It replaces, at the suggestion of an external review, a restatement of the Lemke Oliver–Soundararajan mod-3 law that formerly occupied this slot and carried no new content (their conjecture remains cited at Conjecture 16’s family level and its verification data— $\hat{c}$ drifting $0.400 \rightarrow 0.370$ over $10^7 \rightarrow 4 \cdot 10^9$ with $(1, 1)/(2, 2)$ symmetry 0.99989—remain in the repository). Verification: over 500 log-spaced samples $n \leq 10^8$, mean $D = 76.5$ against mean $D_{\text{sys}} = 64.5$ (ratio 1.19, sampling noise ≈ 0.2), the $(3, 3)$ -lane lead significant at 5.2 standard errors of the mean; per- n sign fraction 0.592, weak exactly as the noise analysis requires.

7 The adversarial audit

Each statement above was attacked three ways.

Layer 1: literature. Five independent literature searches combed OEIS, arXiv, and the standard references for prior art, producing the novelty labels and the attribution section below. This layer also falsified one historical claim in our working notes (the $k \geq 1$ Stern list of Conjecture 14 was described as partly new; it is OEIS A060003 verbatim, and 1493 is the largest known Stern prime); the notes were corrected on the record.

Layer 2: scripted refutation. Every falsifiable-by-instance statement was pushed at least a decade past its original bound: exception hunts across $(10^8, 10^9]$ for Conjectures 13, 14, 15, 17; the uniformity of Conjecture 16 stressed on $d \leq 6000$; the statistical laws re-tested at $4 \cdot 10^9$ (Conjectures 18–21, 25, and the quintuplet count of Conjecture 4); the recovery clause of Conjecture 22 tested on fresh moduli $q \in (3000, 6000]$. Outcome: one kill (Finding 2); no other statement moved.

Layer 3: clean-context replication. Independent replications, working from the bare statements alone without access to our code or notes, recomputed constants from definitions and recounted sequences with their own implementations. All recomputed constants landed inside the stated truncation error; all recounts agreed (one replication initially disagreed at $z \approx -22$, traced by the replicator itself to a bug in *its* sieve—every prime $n \leq Q$ had been struck out by its own modulus—after which it agreed at $z < 1.6$). The same layer reproduced the Fibonacci deficit of Remark after Conjecture 9 and the ordering anomaly of Remark 1, confirming both are properties of the primes rather than of our code.

We draw one methodological conclusion. The only fatal error among twenty-five statements was not statistical but *algebraic*: a factorization living on a density-zero subsequence. Probabilistic sanity checks, however extensive, integrate over density and cannot see such families; only structural attack surfaces (here: “test the claimed exception census on special subsequences”) find them. An audit that only re-runs the original verification at a larger bound would have declared the false version of Conjecture 13 sound.

8 Attribution and novelty

Previously stated in essence, with our contribution being the uniform quantification, computed constants, and re-verification: C1 (OEIS A080149; [28]); C3 [8]; C6 (constant catalogued as OEIS A188596); C9 [5]; C12 [4]; C13-core [1]; C14 (OEIS A060003); C15 [9]; C16-core (uniform Hardy–Littlewood B, cf. [10]); C17 [7]; C18 [11]; C23 [18]; C24-core [1, 19]. (The *retired* former occupants of slots 10 and 25 were previously stated—Caldwell–Gallot [4] and Lemke Oliver–Soundararajan [20] respectively; the third external review caught a stale attribution table that

still listed the current occupants under those citations, and the inconsistency is corrected here: the primordial-twin and lane-race statements now in those slots are ours, with the ingredient heuristics attributed above.)

For the remainder we distinguish, following the external review, two grades of novelty. *Bibliographic novelty only*—unrecorded specializations of Bateman–Horn, analogous to evaluating a classical special function at a new argument, offered as benchmark data: the constants of C2, C5, C7 and C8. *Conceptual novelty*—a mechanism, law, or dichotomy we could not find in the literature: C4’s power-obstruction ladder; C10’s primordial-twin finiteness (the convergent Borel–Cantelli exemplar); C11’s candidate local-global product for a mixed polynomial–exponential sequence; C19’s liminf constant (against the slope-1 laws of Shanks, Wolf, and Kourbatov–Wolf [21]); C20’s microscopic variance law (a sharpening of [15], with the mesoscopic numerics of [22] as precedent); C21’s race mechanism (against the twin-bias literature [23, 24]); C22(ii)’s deficit decomposition; and C25’s Goldbach lane race. Among these we consider C21, C22(ii) and C25—the two square-contamination races and the least-prime deficit law with its ordering anomaly—the paper’s most genuinely new content, and Question 1 its best export.

Question 1. Determine $\theta(q)$ of Conjecture 22: derive, from the Hardy–Littlewood correlations or otherwise, the first-order deficit of $\mathbb{E}_a \text{Li}(p(a, q))/\varphi(q)$ below its random-model value 1—equivalently, explain quantitatively why the primes in their natural order occupy residue classes faster than any exchangeable resampling of themselves.

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